

BLINKIT

Sales & Outlet Performance Analysis Professional Data Analytics Project Report

Project Type	Exploratory Data Analysis & Business Intelligence
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI
Dataset Size	8,523 records × 12 fields
Core Business Metric	Sales performance across products and outlets

1. Executive Summary

This project analyzes Blinkit-style retail sales data to understand overall sales performance, product-category contribution, fat-content contribution, outlet characteristics, outlet location tiers, and establishment-year patterns. The analytical workflow begins in Python for data inspection, cleaning, aggregation, and visualization, and is complemented by an interactive Power BI dashboard for business-facing reporting.

The dataset contains 8,523 records and 12 fields. The Python analysis reports total sales of \$1,201,681, average sales of approximately \$141, and 1,559 unique item identifiers. The Power BI dashboard reports the same overall sales at approximately \$1.20M, an average sales value of \$141, 8,523 sales records, and an average rating of 3.92. The Python notebook rounds average rating to 4.0, so the difference is a display-precision issue rather than a substantive contradiction.

Key findings

- Low Fat products contribute 64.6% of total sales, compared with 35.4% from Regular products.
- Fruits and Vegetables and Snack Foods are the two largest item-type contributors, at about \$178K and \$175K respectively.
- Tier 3 outlets generate the highest sales among the three location tiers, at approximately \$472.13K.
- Medium-sized outlets account for the largest share of sales at 42.3%, followed by Small outlets at 37.0%.
- Supermarket Type 1 is the dominant outlet type in the dashboard, contributing about \$787.55K in sales.
- The establishment-year analysis shows a strong 1998 sales value, followed by a relatively stable band around \$129K-\$133K for many later years, with a notable dip in 2011.

Source basis: supplied Python project report and Power BI dashboard. The Python notebook documents the data structure, cleaning, KPI calculations and six analytical charts. ■filecite■turn0file0■L42-L73■

2. Business Problem & Objectives

The project is designed to convert raw outlet-level product sales data into actionable business information. The central analytical question is: which products, product characteristics, outlet characteristics and locations are associated with stronger sales performance?

Project objectives

- Measure total sales and average sales performance.
- Understand the contribution of Low Fat and Regular products.
- Compare sales across item types and identify high- and low-contributing categories.
- Compare fat-content sales across Tier 1, Tier 2 and Tier 3 outlet locations.
- Examine total sales by outlet establishment year.
- Assess the contribution of outlet size and outlet type.
- Present the results in a management-friendly Power BI dashboard.

The supplied notebook explicitly defines these analytical requirements through KPI and chart objectives, including fat content, item type, city tier, establishment year, outlet size and location analysis. ■filecite■turn0file0■L78-L106■

3. Dataset & Methodology

Dataset overview

Attribute	Description
Rows	8,523
Columns	12
Primary measure	Sales
Product fields	Item Fat Content, Item Identifier, Item Type, Item Weight, Item Visibility
Outlet fields	Outlet Identifier, Outlet Establishment Year, Outlet Location Type, Outlet Size, Outlet Type
Customer/product quality field	Rating

Data types

The source analysis identifies categorical fields such as Item Fat Content, Item Identifier, Item Type, Outlet Identifier, Outlet Location Type, Outlet Size and Outlet Type. Outlet Establishment Year is integer, while Item Visibility, Item Weight, Sales and Rating are numeric floating-point fields. ■filecite■turn0file0■L45-L66■

Analytical workflow

Stage	Activity
1. Data ingestion	Load the CSV dataset into a Pandas DataFrame.
2. Profiling	Inspect shape, columns and data types.
3. Cleaning	Standardize inconsistent Item Fat Content labels.
4. KPI creation	Calculate total sales, average sales, unique items and average rating.
5. Segmentation	Group sales by fat content, item type, outlet tier, establishment year, outlet size and location.
6. Visualization	Create donut, bar, grouped bar, line and horizontal bar charts in Python.
7. BI reporting	Present KPIs and visual analysis in Power BI.

4. Data Preparation & Cleaning

The project performs a focused cleaning step on the Item Fat Content field. The raw data contains multiple representations of the same two business categories: 'Regular', 'Low Fat', 'low fat', 'LF' and 'reg'. These values were standardized so that 'LF' and 'low fat' become 'Low Fat', while 'reg' becomes 'Regular'.

After cleaning, the notebook confirms that the field contains only two standardized values: Low Fat and Regular. This step is important because grouping before standardization would split the same business category across multiple labels. The exact transformation is documented in the supplied notebook. ■filecite■turn0file0■L67-L77■

Data quality note

The supplied materials document the categorical standardization above, but they do not provide a full missing-value, duplicate-record, outlier or statistical-validity audit. Therefore, this report does not claim that those checks were completed.

Recommended production-grade validation

- Quantify missing values for every column before and after cleaning.
- Check duplicate Item Identifier + Outlet Identifier combinations where appropriate.
- Inspect extreme values for Sales, Item Weight and Item Visibility.
- Validate whether Rating is on a fixed 1-5 scale and investigate unusual values.
- Document assumptions for any imputation or removal of records.

5. KPI Overview

KPI	Value	Interpretation
Total Sales	\$1,201,681	Overall sales value in the Python analysis.
Average Sales	\$141	Average sales value per record, rounded in the source.
Unique Items	1,559	Distinct Item Identifier values in the Python analysis.
Average Rating	3.92	Power BI dashboard value; Python displays 4.0 after rounding.
Sales Records	8,523	Number of records shown in the Power BI dashboard.

The Python notebook calculates total sales, average sales, number of distinct item identifiers and average rating directly from the DataFrame. It reports \$1,201,681 total sales, \$141 average sales, 1,559 unique items and an average rating of 4.0 to one decimal place. The Power BI dashboard shows \$1.20M total sales, \$141 average sales, 8,523 records and 3.92 average rating. [filecite turn0file0L78-L95](#) [filecite turn0file1L97-L102](#)

6. Product & Fat-Content Analysis

6.1 Sales by Fat Content

Low Fat products account for 64.6% of total sales, with \$776,319.68 in sales, while Regular products account for 35.4%, with \$425,361.80. Average sales per record are very close: \$140.71 for Low Fat and \$141.50 for Regular. This indicates that the large difference in total sales is primarily associated with the distribution/volume of items rather than a large difference in average sales value.

Fat Content	Total Sales	Share	Avg. Sales	Unique Items
Low Fat	\$776,319.68	64.6%	\$140.71	1,008
Regular	\$425,361.80	35.4%	\$141.50	551

The source notebook's donut chart also visualizes the same 64.6% versus 35.4% split.

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6.2 Sales by Item Type

Fruits and Vegetables is the largest item-type contributor at approximately \$178,124, followed closely by Snack Foods at approximately \$175,434. Household, Frozen Foods and Dairy form the next major contributors. Seafood is the smallest category in the supplied analysis at approximately \$9,078.

Rank	Item Type	Total Sales
1	Fruits and Vegetables	\$178,124
2	Snack Foods	\$175,434
3	Household	\$135,977
4	Frozen Foods	\$118,559
5	Dairy	\$101,276
6	Canned	\$90,707
7	Baking Goods	\$81,895
8	Health and Hygiene	\$68,026
9	Meat	\$59,450
10	Soft Drinks	\$58,514
11	Breads	\$35,379
12	Hard Drinks	\$29,335
13	Others	\$22,452
14	Starchy Foods	\$21,880
15	Breakfast	\$15,597
16	Seafood	\$9,078

The notebook's item-type visualization is explicitly intended to identify category performance using total sales, while also calculating average sales, number of unique items and average rating for each item type.

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7. Outlet & Geographic Analysis

7.1 Fat Content Across Location Tiers

Tier 3 outlets generate the highest total sales at approximately \$472,133, followed by Tier 2 at \$393,151 and Tier 1 at \$336,398. Within every tier, Low Fat sales exceed Regular sales. The Tier 3 Low Fat segment is the largest single fat-content/tier combination in the supplied analysis.

Location Tier	Low Fat Sales	Regular Sales	Total Sales
Tier 1	\$215,048	\$121,350	\$336,398
Tier 2	\$254,465	\$138,686	\$393,151
Tier 3	\$306,807	\$165,326	\$472,133

The source notebook defines this chart specifically to compare total sales across outlets segmented by fat content and location tier. `filecite[turn0file0]L175-L205`

7.2 Sales by Outlet Size

Medium outlets contribute the largest share of sales at 42.3%, followed by Small outlets at 37.0% and High-sized outlets at 20.7%. The analysis therefore shows that the Medium outlet segment is the largest contributor in the dataset.

Outlet Size	Sales Share
Medium	42.3%
Small	37.0%
High	20.7%

The notebook groups sales by Outlet Size and visualizes the resulting distribution. `filecite[turn0file0]L240-L261`

7.3 Sales by Outlet Location Type

Outlet Location Type	Total Sales
Tier 3	\$472.13K
Tier 2	\$393.15K
Tier 1	\$336.40K

The Power BI dashboard presents the same location pattern visually, with Tier 3 at \$472.13K, Tier 2 at \$393.15K and Tier 1 at \$336.40K. `filecite[turn0file1]L53-L69`

8. Outlet Establishment Analysis

Sales vary materially by outlet establishment year. The highest value shown in the supplied Python chart is approximately \$204,522 for the 1998 establishment year. A notable low point appears in 2011 at approximately \$78,131. Most of the later observations sit in a relatively narrow range around \$129K-\$133K.

Establishment Year	Total Sales
1998	\$204,522
2000	\$131,809
2007	\$131,113
2010	\$132,113
2011	\$78,131
2012	\$130,476
2015	\$130,942
2017	\$133,103
2020	\$129,103
2022	\$131,477

The source notebook sorts establishment years chronologically and plots total sales as a line chart with labels for each point. The Power BI dashboard also contains an establishment-year visual showing the same broad pattern.

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Interpretation

- The 1998 observation is materially higher than the other establishment-year values in the supplied chart.
- The 2011 dip should be investigated before treating it as a business trend, because the source materials do not establish the underlying cause.
- The relative stability of many later years suggests that establishment year alone does not explain most of the variation in sales in this dataset.

9. Power BI Dashboard

The Power BI dashboard converts the analysis into a compact management view. It combines headline KPIs with visuals for establishment year, fat content, item type, outlet size, outlet location type and outlet type. The dashboard reports \$1.20M total sales, \$141 average sales, 8,523 sales records and a 3.92 average rating.

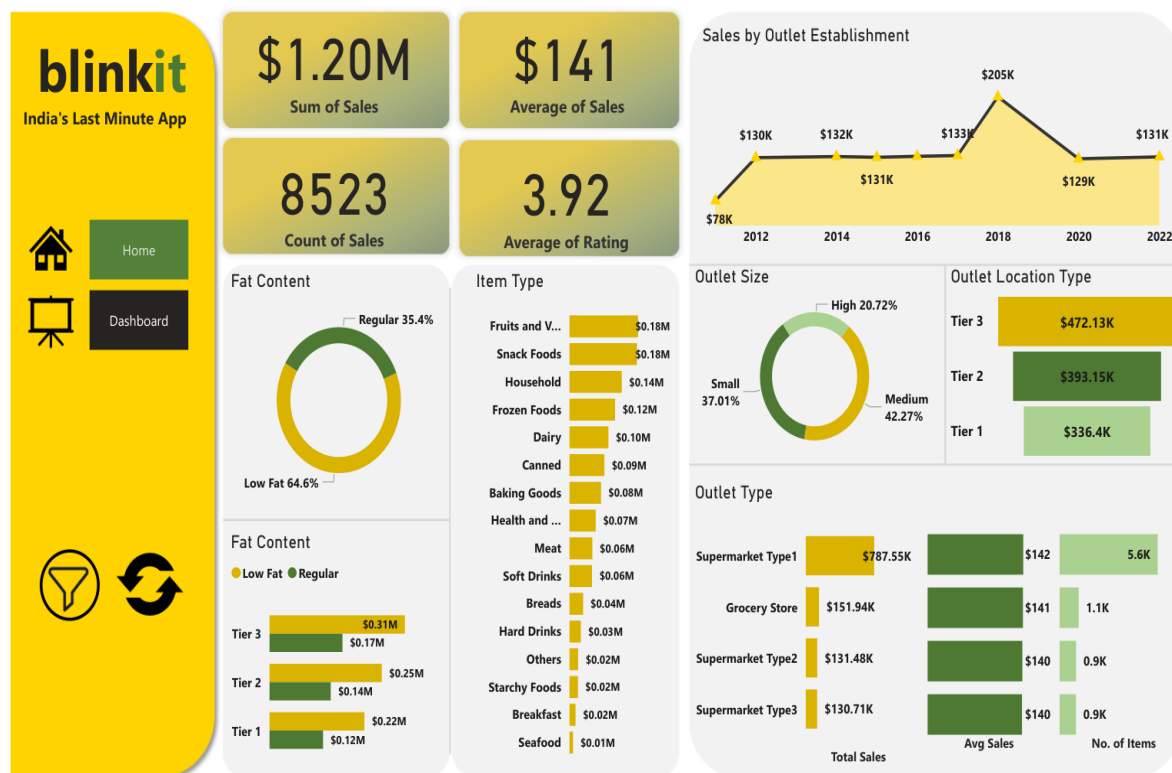


Figure 1. Supplied Power BI dashboard, page 2 of the dashboard PDF.

Dashboard observations

- The dashboard places the most important KPIs at the top, enabling quick performance assessment.
- The item-type visual highlights Fruits and Vegetables and Snack Foods as the leading categories.
- Outlet Type analysis shows Supermarket Type 1 as the dominant sales contributor at about \$787.55K.
- The dashboard uses multiple visual encodings to compare outlet size, location tier and fat content.

The dashboard source lists the item-type, outlet-size, fat-content, location, establishment-year and outlet-type values used in these observations. ■filecite■turn0file1■L8-L102■

10. Technical Implementation

Python layer

The project uses NumPy, Pandas, Matplotlib and Seaborn. Pandas is used for ingestion, profiling, cleaning, grouping and KPI calculation; Matplotlib and Seaborn are used for visualization. The source notebook explicitly imports these libraries before loading the CSV file. [filecite turn0file0L2-L10](#)

Core analytical techniques

Technique	Purpose in project
groupby()	Segment sales by product and outlet dimensions.
agg()	Calculate sums, means and unique-item counts.
replace()	Standardize inconsistent fat-content labels.
nunique()	Count distinct item identifiers.
Matplotlib / Seaborn	Produce business-focused charts.
Power BI	Build a consolidated dashboard for decision support.

Reproducibility

The Python notebook saves several visual outputs as PNG files at 300 DPI, including the fat-content, item-type, city-tier, establishment-year, outlet-size and location visuals. This provides a useful reproducibility trail for the analytical workflow.

11. Business Insights & Recommendations

Key insights

Product mix: Low Fat products generate the majority of sales, contributing 64.6% of total sales.

Category priorities: Fruits and Vegetables and Snack Foods are the two largest item-type contributors.

Location performance: Tier 3 outlets lead sales among the three location tiers.

Outlet size: Medium outlets represent the largest outlet-size sales share at 42.3%.

Outlet format: Supermarket Type 1 is the largest outlet-type contributor in the dashboard.

Year pattern: The establishment-year series contains a high 1998 value and a notable 2011 low point, with many later years clustered around \$129K-\$133K.

Recommended actions

- Prioritize inventory and promotional planning around high-contributing categories, especially Fruits and Vegetables and Snack Foods.
- Use the strong Low Fat contribution as an input into assortment planning, while continuing to monitor Regular-product performance because average sales are very similar.
- Investigate Tier 3 outlet characteristics to understand why they generate the highest sales and whether the model can be replicated elsewhere.
- Review the 2011 establishment-year anomaly and the unusually high 1998 value before making operational decisions based on outlet age.
- Use the Power BI dashboard as a recurring management view, with additional filters for item type, outlet type, tier and fat content if the underlying PBIX supports them.
- For a stronger business case, add margin, quantity, discount, inventory and time-based order fields in a future version so sales performance can be connected to profitability and operational efficiency.

12. Limitations & Future Scope

Limitations

- The supplied project focuses on sales and descriptive segmentation; it does not establish causality.
- No profit, cost, discount, quantity or inventory measures are included in the supplied fields, so profitability and stock-efficiency cannot be evaluated.
- The source materials document a targeted cleaning step but do not document a complete missing-data and outlier treatment workflow.
- The establishment-year analysis contains a relatively small set of year values, so year-level differences should be interpreted carefully.

Future enhancements

- Add time-series analysis with monthly/weekly sales if transaction dates become available.
- Introduce profitability KPIs such as gross margin and contribution margin.
- Build drill-through pages for item-level and outlet-level performance.
- Add slicers for Item Type, Fat Content, Outlet Type, Outlet Size and Location Tier.
- Create automated refresh and a documented data pipeline for repeatable reporting.
- Apply statistical testing or predictive modeling to identify drivers of sales rather than only describing associations.

Conclusion

The Blinkit analysis successfully demonstrates an end-to-end descriptive analytics workflow: raw data ingestion, targeted cleaning, KPI calculation, segmentation, visualization and business dashboarding. The results show a sales mix led by Low Fat products, strong contributions from Fruits and Vegetables and Snack Foods, higher sales from Tier 3 locations, and a dominant contribution from Supermarket Type 1 outlets. The project is suitable as a portfolio-level demonstration of practical Python and Power BI analytics, while the recommended enhancements would make it more robust for production-style business analysis.

Source note: All numerical findings in this report are derived from the two supplied project PDFs. Where the Python and Power BI materials display different rounding or aggregation labels, the report preserves the source values and explicitly notes the difference.

Appendix A. Source Project Visual

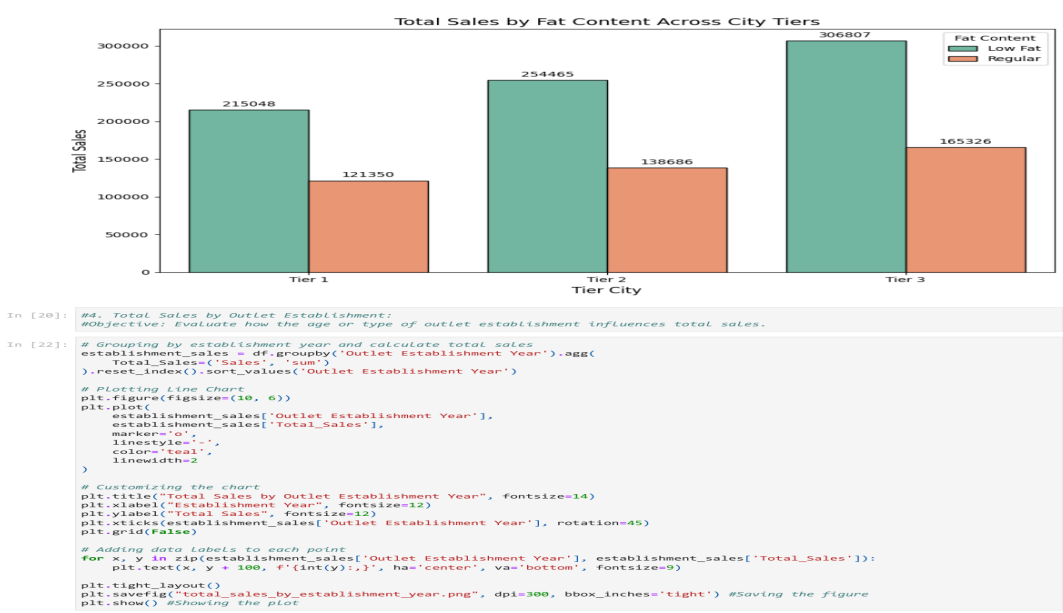


Figure 2. Supplied Python analysis page showing outlet-size and establishment-year visualizations.

Appendix B. Reported Field List

Field	Type in source analysis
Item Fat Content	object / categorical
Item Identifier	object / categorical
Item Type	object / categorical
Outlet Establishment Year	int64
Outlet Identifier	object / categorical
Outlet Location Type	object / categorical
Outlet Size	object / categorical
Outlet Type	object / categorical
Item Visibility	float64
Item Weight	float64
Sales	float64
Rating	float64